MindMemory

COMINDLAB POSITION PAPER · CORPORATE EDITION

Memory at Work.

Why the next productivity gain in knowledge work won't come from smarter AI. It will come from AI that *remembers you*.

CoMindLab Labs · April 2026 · Author: Igal Geron, co-founder

01 The unspoken cost of AI today

Every senior leader your organisation employs has the same complaint about AI tools: **"I have to keep telling it the same things."**

The marketing director re-explains brand voice every chat. The CFO re-pastes last quarter's structure into every analysis. The CHRO re-teaches what "high potential" means in your culture, every single time. The work is real. The AI is capable. The friction sits in between — in the absence of memory.

This isn't a prompt-engineering problem. It's a **context-engineering** problem. And it's the single biggest lever left for making AI tools feel less like a search box and more like a colleague.

> **The thesis:** AI capability has plateaued at "smart but forgetful." The next 10× isn't a better model. It's a structured, persistent memory that turns every correction into a permanent improvement.

02 What "memory" actually means at work

Most AI products that mention memory mean something narrow: chat history. The system remembers what you said earlier in *this* conversation. That's session state, not memory. Close the tab and it's gone.

Real memory has four properties of a competent colleague who has worked with you for two years:

IDENTITY

A stable sense of who they are and how they work — not re-set every Monday.

LONG-TERM

Confirmed lessons that persist for months. "We tried that in 2024, here's why it didn't

work."

WORKING

This week's focus, this month's priorities. Knows what you're in the middle of.

ARCHIVE

Old context that's not loaded by default but retrievable when relevant.

This is how a strong analyst, EA, or trusted peer carries you across years of work. It's also what AI tools have been missing — and what we've now built for the case where the work itself is computational: software, analysis, content, decision support.

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03 MindMemory — the visual model

We built MindMemory as a research preview to test one question: **can a structured memory layer make an off-the-shelf AI tool measurably more useful, week after week?**

The visual on the next page summarises the architecture. Four panels:

- **From agents to minds** — the evolution from stateless tools to something that behaves like a colleague.
- **Four tiers** — how memory is organised by lifespan, with a promotion ladder from observation to permanent rule.
- **Three concentric loops** — how the system curates itself *continuously*, not just at the end of a session.
- **Findings** — three months of measured data, including one honest null result.

MindMemory Memory for Claude Code that gets better the more you use it.

PANEL A · CONCEPT

From agents to minds.

Capability has plateaued; **memory is the next lever**. A "mind" is identity + memory + heuristics, wired together as one markdown construct.

01

Agents

`stateless`

- High capability per call
- No memory
- No personality
- Transactional

02

Claude Code today

`session-scoped`

- High capability
- Session-only memory
- Thin prompted personality
- Transactional

MINDMEMORY

03

A mind

`colleague-like`

- High capability
- Persistent memory
- Stable identity
- Growing heuristics

04

Human mind

`aspiration`

- Variable capability
- Lifelong, selective
- Deep, stable
- Continuous learning

A mind is one markdown-based construct: `identity` + `memory` + `heuristics`. **Claude Code is the substrate. Markdown is the storage.** No vector DB, no fine-tuning.

PANEL B · MEMORY

Four tiers, by lifespan.

Organised by how long things live, not by topic. Index pattern keeps always-loaded context to ~1–2 KB while making thousands of memories available on demand.

B1 · The 4-tier stack

TOP = LONGER-LIVED

01 Identity — READ-ONLY
Personality. Rules it never breaks.
`identity/anchor.md`

02 Long-term memory — WEEKS+
Confirmed patterns, profile, lessons.
`memory/MEMORY.md · {topic}.md`

03 Working memory — DAYS
This week — focus, blockers, recent corrections.
`memory/working-memory.md`

04 Archive — RARELY READ
Compressed history. Not loaded by default.
`memory/archive.md`

B2 · Hunch → heuristic

3-CONFIRMATION RULE

1× **Observation** `logged`
Noted by collector. Not yet in memory.

2× **Hunch** `tentative`
Added to working-memory, labelled.

3× **Heuristic** `permanent`
Promoted to Tier 2. Applied going forward.

↺ **Contradicted** `demoted`
Returns to hunch, or archived with note.

Each Tier-2 entry stores **why** + **how to apply** — so the mind can judge edge cases instead of blindly following.

PANEL C · SELF-MAINTENANCE

Three concentric loops.

The mind curates itself **while you work**, not just at logoff. Catching corrections within minutes produces tighter rules than catching them hours later.

INNER LOOP — Mid-session — every N tool calls

PostToolUse hook

Collect signals — `memory-collector.py`
Every **~10 calls**: scans live transcript, appends to `working-memory.md` while context is hot.
- correction
- confirmation
- follow-through
- frustration

→

Promote / demote — `memory-curator.py`
Every **~25 calls**: applies the 3-confirmation rule. Hunch → heuristic. Contradicted → demoted. Stale (>5d) → archived.

MIDDLE LOOP — Per session — bookends

SessionStart · SessionEnd

Session start — `session-start-health.py`
Loads **identity** + **MEMORY.md** + **working-memory**. Warns if curator hasn't run in >7 days.

→

Session end — `session-metrics.py`
Writes one row to `session-metrics.csv`. Heavy maintenance already happened in the inner loop — this is the safety net.

OUTER LOOP — Weekly — time-gated

7-day timestamp

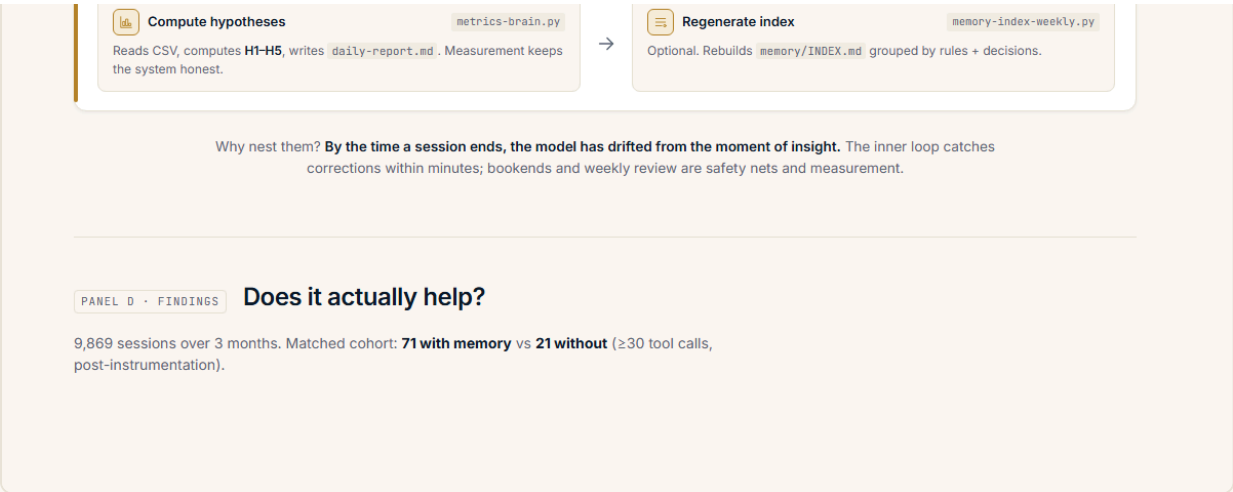

Figure 1. The MindMemory architecture. Single-page reference of the system in production at CoMindLab.

04 What the data says — three months in

We instrumented every session over a 12-week period. 9,869 sessions captured. After applying matching criteria (≥30 tool calls, post-instrumentation cutoff), 92 sessions form the comparable cohort: 71 with memory loaded vs 21 without.

PRODUCTION	ORIENT	DENSITY	REPEAT
2.2×	−10%	4.3×	Null
18.5% vs 8.5%	14.6 vs 16.3 calls	26.0 vs 6.0 edits	primary outcome

What the wins say. When memory is loaded, sessions are denser, more productive, and orient to real work faster. The single largest effect — 4× more productive output per session — is consistent with the qualitative report from the primary user: "*You remembering things helps me. I am much less frustrated with the process.*"

What the null says. The headline outcome — fewer repeated corrections — has not yet been demonstrated. Three honest theories: selection bias (memory loads on harder work); coverage gap (memory doesn't yet hold the right lessons); recall gap (memory is there but not applied under pressure). We're publishing the result rather than burying it. The system measures itself; we report what the measurement says.

> **The honest framing:** 3× more useful output per session. That's not a marketing number — it's what the instrumented data shows. The repeat-correction outcome remains an open question.

05 Why this matters for your organisation

If you employ knowledge workers and they use AI tools daily, this question is on your desk whether you've named it or not:

Every correction your team types into an AI tool is a piece of organisational knowledge that vanishes the moment they close the tab.

Across thousands of sessions per week per company, the cumulative leak is enormous. Brand voice, regulatory context, customer history, internal vocabulary, lessons learned — all retyped, every time, by the people whose time is most expensive.

MindMemory is one answer. The pattern is general:

- **Capture corrections automatically.** Don't ask people to write FAQs nobody reads.
- **Promote what gets confirmed.** A lesson seen three times becomes a rule. A rule contradicted gets demoted. The system curates itself.
- **Measure whether memory helps.** If you can't measure it, you can't improve it. This applies to every AI tool your organisation deploys.

The CoMindLab thesis: **the 5% of organisations that succeed with AI don't have better technology. They have better collaboration design.** Memory is one of the highest-leverage design choices available right now, and it's being left on the table by tools that focus on capability instead of continuity.

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06 What we're offering

MindMemory itself is open source (MIT) and free for any team to install. The repository is at github.com/CoMindLab-ai/mind-memory.

For organisations that want the pattern adapted to their specific work — brand voice, regulatory context, decision frameworks, role-specific memory layers — CoMindLab offers three engagement models:

CO-DEVELOPMENT · 1–3 WEEKS

One team, one problem. We embed, build a memory layer for your highest-friction AI workflow, and leave it running.

HYPER-CUSTOMISED · 3–8 WEEKS

Memory layers across multiple roles. Identity templates per function. Measurement harness adapted to your KPIs.

AI-NATIVE PROCESS · 6–12 WEEKS

Memory + identity + heuristics designed into a full operating model. For teams treating AI as colleague-grade infrastructure, not a side tool.

WORKSHOP · 1 DAY

Hands-on session for your leadership team: design a memory layer for one strategic AI use case in your business, leave with a working prototype.

07 How to engage

Three concrete next steps, in order of commitment:

- **Read the research.** The full repository, including the architecture, the findings, and the open questions, is public. Two hours of reading will tell you whether this matches a real problem in your organisation.
- **Try it.** Install the starter kit on one Claude Code project. Use it for two weeks. The measurement harness will tell you whether memory is helping *your* work.
- **Talk to us.** If the pattern resonates and the friction it addresses sits squarely on your team's calendar, write to home@comindlab.ai. We respond personally.

> **The bet:** in 24 months, every serious knowledge organisation will have a memory layer between its people and its AI tools. The only question is whether you build yours deliberately, or accept whatever the platforms eventually impose.